

EN.553.443/633 Monte Carlo Methods

Spring 2026 Syllabus

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1 Course Information

Course: EN.553.443/633 Monte Carlo Methods, Spring 2026

Instructor: Apurva Nakade, apurva.nakade@jhu.edu

Office hours and TA information will be posted on Canvas.

1.1 Class Meetings

Class Time	Location	Discussion Time	Location
MW 01:30 PM – 02:45 PM	Hodson 313	T 06:00 PM – 06:50 PM	Maryland 201

Please check SIS for any updates to the schedule.

1.2 Textbook

Online Course Notes: <https://apurvanakade.github.io/Monte-Carlo-Methods/>

Recommended Textbooks:

- Rubinstein, R. Y. and Kroese, D. P. (2017), *Simulation and the Monte Carlo Method* (3rd edition), Wiley, ISBN: 978-1-118-63216-1. Available online with a JHU account or in print form.
- Dobrow, R. P., *Introduction to Stochastic Processes With R*, Wiley, DOI: 10.1002/9781118740712. Available on Wiley Online Library for free using JHU credentials.

Other resources will be posted on the course website as needed.

1.3 Prerequisites

Students should come into the course with a good working knowledge of linear algebra and of probability and some statistics at the advanced undergraduate level. Students should also have some computer programming experience in Python and will be expected to submit programming assignments in Jupyter notebooks.

1.4 Course Description

Computer simulation and related Monte Carlo methods are widely used in engineering and scientific work. Simulation provides a powerful tool for the analysis of real-world systems when they are not amenable to traditional analytical approaches. This course introduces upper-level undergraduates and graduate students to fundamental tools for designing, conducting, and interpreting Monte Carlo studies.

This is a rigorous mathematics course with strong mathematical expectations from students. The entire focus is on the mathematical analysis of algorithms, including their theoretical foundations, convergence properties, and error analysis. While programming assignments are an essential component of the course, this is not a programming or software engineering course. Similarly, we will not cover applications of these algorithms to specific fields such as finance, physics, or machine learning. Instead, we emphasize generic mathematical principles and rigorous theoretical understanding that underlie Monte Carlo methods.

As an applied mathematics course, the emphasis is on generic principles and algorithms that are widely applicable in simulation, rather than on detailed discussion of specific applications or software packages. Upon completion of this course, students will have the insight and understanding to critically evaluate and use many state-of-the-art methods in simulation.

2 Grading

The following is the breakdown of the grading scheme:

Component	Percentage
Weekly Homework	15%
Weekly Quizzes	60%
Final Project	25%

At the end of the semester, grades will be assigned such that the median grade is at least B+. Grade assignments will generally follow university standards but are ultimately at the discretion of the instructor. Final grades are not based solely on arbitrary percentage cutoffs but also reflect your overall learning, improvement over the semester, engagement with course material, and **participation in class discussions**.

2.1 Quizzes

There will be short weekly quizzes during discussion sections. The quizzes will be based on the homework and lectures. The quizzes will be closed book. Only basic function calculators will be allowed during a quiz. Quizzes will focus on the material covered in the previous week.

2.1.1 Grade Adjustments

Your lowest homework score will be dropped automatically at the end of the semester.

For quizzes, instead of dropping your lowest score, at the end of the semester your overall quiz grade will be multiplied by $n/(n-1)$, where n is the total number of quizzes (up to a maximum of 100%). This adjustment is mathematically equivalent to dropping one quiz but disincentivizes intentionally missing a quiz.

In case of a medical emergency, illness, or school/work-related absence, you must:

- Promptly inform the instructor,
- Provide appropriate documentation (e.g., doctor's note, official travel documentation, university-approved absence form).

Decisions on excused absences will be made on a case-by-case basis. If approved, you will be required to complete a make-up quiz as an *oral examination* conducted by the instructor.

2.2 Homework

Homework will be due weekly. Late homework will not be accepted. Homework will consist of both theoretical and programming problems. The programming problems will be done using Jupyter notebooks.

2.3 Final Project

The final project will be done in **groups of 3 students**. You'll need to make a 10 minute presentation and submit a report. The project will be graded based on the quality of the presentation, the report, and the code.

For the project, you'll have to implement an algorithm that we didn't cover in class possibly from the textbook or from a research paper. More details will be provided later in the semester.

The final project presentations will be held during the **final exam slot** for this course. **Attendance is mandatory** for all students. Remote participation will not be permitted, and failure to attend will result in loss of all presentation credit.

3 Expectations from students

Please review the university-wide policies and resources provided on the [Canvas Course Syllabus](#) page.

3.1 Deadlines

You are expected to submit homework by each deadline. Late submissions will not be accepted for any reason. Each homework will be posted a week in advance and there is no reason to wait until the last minute to submit it.

3.2 Attendance and Classroom Policies

Regular attendance is expected and is essential for success in this course. Students should arrive before class begins. Arriving late is disruptive to the learning environment. Students who repeatedly arrive late will not be admitted to class and will need to meet with the instructor to discuss their situation.

Exception: Students who have another class or work commitment that conflicts with the start time and prevents them from arriving on time should inform the instructor well in advance to receive an exception.

Laptops and phones are not allowed in class. Tablets are allowed for note-taking only. Research has shown that handwritten notes lead to better retention and understanding of material compared to typed notes. Additionally, devices can be distracting both to the user and to other students in the classroom. Violations of this policy will result in exclusion from class.

3.3 Generative Artificial Intelligence (AI) Tools

Use of generative artificial intelligence (AI) tools such as ChatGPT can augment learning experiences when used appropriately. You are encouraged to use generative AI to brainstorm and generate code, refine ideas, find reliable sources, outline, check grammar, and format bibliographies. You should note, however, that the material generated by these programs may be inaccurate, incomplete, biased, or otherwise problematic. You are ultimately responsible for what you submit.

All writing and calculations you submit must be your own. Use of AI to solve math problems is strictly prohibited and is detrimental to your learning. This would be a violation of the academic integrity policy and appropriate action will be taken. You will not learn anything if you do that and it will be reflected in your quiz scores, which are worth 60% of your grade. Use your interaction with AI as a learning experience. Then, let your submitted work reflect your improved understanding.

Good Uses of AI Tools

3.3.1 Good Uses of AI Tools

- Asking how to perform standard tasks such as plotting, statistical testing, or finding Python functions
- Getting help with debugging error messages or code optimization
- Generating small pieces of code
- Generating ideas for project topics or alternative approaches
- Learning syntax, documentation, or programming concepts
- Assistance with grammar, formatting, or report structure

3.3.2 Bad Uses of AI Tools

- Pasting entire homework assignments and asking it to solve them
- Using it to generate large chunks of code without understanding the logic
- Having AI solve mathematical proofs or theoretical problems
- Copy-pasting AI solutions without modification or understanding
- Using AI as a substitute for reading course materials or attending lectures